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EasyVisa

ROBUST DATA MODELING WITH ENSEMBLING AND TUNING TECHNIQUES

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1. Introduction

In the modern globalized economy, organizations increasingly rely on international talent to address skill gaps, enhance innovation, and maintain competitive advantage. Companies across industries such as information technology, engineering, healthcare, finance, and research frequently recruit highly skilled professionals from different countries to support business expansion and operational efficiency. However, hiring foreign employees involves navigating complex regulatory frameworks and visa approval processes that require careful evaluation of eligibility criteria. These criteria often include factors such as educational qualifications, job experience, offered wages, employment conditions, employer credibility, and compliance with labour regulations. The complexity and volume of visa applications make the decision-making process time-consuming and resource-intensive for both organizations and regulatory authorities. Traditionally, visa approval decisions have been based on manual review processes, where applications are evaluated individually against predefined rules and documentation requirements. While such methods ensure compliance with legal standards, they may also lead to inconsistencies, delays, and inefficiencies when dealing with large datasets. As organizations continue to expand globally and the number of applications increases, there is a growing need for intelligent systems that can assist in evaluating applications more efficiently and accurately. Data-driven decision-making has therefore emerged as an essential approach to improve the reliability, speed, and transparency of visa approval processes.

The advancement of data analytics and machine learning technologies has created new opportunities for organizations to extract meaningful insights from historical data. Machine learning models are capable of identifying hidden patterns, trends, and relationships within large datasets that may not be immediately visible through traditional analysis. By leveraging these techniques, organizations can predict outcomes based on past observations, thereby improving the consistency and effectiveness of decision-making processes. Predictive analytics not only reduces human effort but also helps organizations allocate resources more efficiently, minimize risks, and enhance strategic planning. This project focuses on developing a predictive model that can analyze historical visa application data and estimate the likelihood of visa approval or certification. The dataset used in the analysis includes multiple variables that influence visa decisions, such as applicant education level, job role, industry sector, required experience, wage levels, working conditions, employer size, and geographical factors. These variables collectively contribute to determining whether a visa application satisfies regulatory and economic requirements. By systematically studying these factors, the project aims to understand which attributes significantly impact approval outcomes and how they interact with one another.

The project begins with Exploratory Data Analysis (EDA), which involves examining the structure, distribution, and relationships present within the dataset. EDA helps in identifying missing values, detecting outliers, understanding variable distributions, and recognizing patterns that may influence model performance. Proper data preprocessing techniques such as handling missing data, encoding categorical variables, and scaling numerical features are applied to ensure that the dataset is suitable for machine learning algorithms. These preprocessing steps are essential for improving model accuracy and ensuring meaningful predictions. Multiple machine learning classification algorithms are implemented to evaluate their effectiveness in predicting visa approval outcomes. Traditional models such as Logistic Regression, Decision Trees, and other classification techniques provide a baseline understanding of predictive performance. In order to further enhance model accuracy and robustness, advanced ensemble techniques are applied. Ensemble learning methods combine predictions from multiple models to reduce bias and variance, resulting in more stable and reliable outcomes. Hyperparameter tuning techniques are also used to optimize model performance by selecting the best combination of parameters that improve prediction accuracy. The use of ensemble methods and model optimization techniques ensures that the final predictive model is not only accurate but also generalizable to unseen data. This improves the practical applicability of the model in real-world business scenarios where decision-makers require reliable insights. By identifying key factors that influence visa approvals, organizations can better understand the characteristics of successful applications and make more informed hiring decisions. This can significantly reduce processing time, improve candidate screening efficiency, and support strategic workforce planning.

This project demonstrates the practical application of machine learning techniques in solving a real-world business problem related to visa approval prediction. By integrating data analysis, predictive modeling, and optimization techniques, the study highlights how organizations can leverage data-driven insights to enhance decision-making processes. The results of this project contribute to a better understanding of the factors influencing visa approval outcomes and provide a foundation for developing intelligent systems that support efficient and consistent business operations in a global talent environment.

2. Business Problem

2.1 Problem Context

In today's competitive global environment, companies in the United States are constantly looking for skilled employees who can contribute to innovation, productivity, and overall business growth. Many industries, especially technology, engineering, healthcare, and finance, often face shortages of qualified professionals within the domestic workforce. To overcome this gap, organizations hire skilled foreign workers who can bring specialized knowledge and expertise. Hiring international employees not only helps companies meet their workforce demands but also supports innovation and global collaboration. However, hiring foreign workers is not a simple process. Employers must follow regulations established under the Immigration and Nationality Act (INA), which allows companies to hire foreign nationals either temporarily or permanently under specific conditions. These regulations ensure that hiring foreign workers does not negatively affect job opportunities, wages, or working conditions for U.S. workers available for the job role and that the salary offered meets the standard wage level for that particular occupation and location. The Office of Foreign Labor Certification (OFLC) is responsible for reviewing employer applications and granting labor certifications when the required conditions are met. The certification process involves evaluating several factors such as job role requirements, salary offered, educational qualifications, experience level, employer details, and work location. Since each application contains multiple variables that influence the final decision, the evaluation process becomes complex and time-consuming. With the increase in globalization, the number of visa applications has increased significantly over the years. Reviewing each application manually requires a large amount of effort, time, and resources. In many cases, manual decision-making may also lead to inconsistencies due to human error or subjective judgment. Therefore, organizations are increasingly looking for data-driven solutions that can make the evaluation process faster, more accurate, and more consistent.

2.2 Problem Definition

In the fiscal year 2016, the OFLC processed approximately 775,979 employer applications covering nearly 1,699,957 job positions for labor certification. This shows that the number of visa applications is continuously increasing every year. As the volume of applications increases, it becomes difficult to manually review every application efficiently. The traditional approach requires significant administrative effort and may result in delays in decision-making. Since multiple factors influence visa approval decisions, it becomes important to identify which variables play a major role in determining the outcome. Factors such as education level, job experience, wage offered, full-time or part-time employment, employer reputation, and job location may significantly affect whether a visa application is approved or denied. Analysing these factors manually can be challenging, especially when dealing with large datasets.

To solve this problem, a machine learning-based solution can be used to analyze historical visa application data and identify patterns that influence certification outcomes. By using predictive models, it is possible to estimate the probability of visa approval for new applicants. This approach can help organizations and decision-makers focus on applications that have a higher likelihood of approval, reducing time and effort required in the evaluation process. EasyVisa, a data analytics company, has been hired to develop a predictive model that can support the visa certification process. As a data scientist at EasyVisa, the task is to analyze the dataset and build a classification model that can predict whether a visa application is likely to be approved or denied.

2.3 Objectives

The main objective of this project is to develop a Machine Learning model that can predict the likelihood of visa approval based on various applicant and employer-related factors. The specific objectives of the project are:

- To analyze historical visa application data to understand the key factors that influence visa approval decisions.
- To build a classification model that can predict whether a visa application will be certified or denied.
- To identify the most important variables that affect the outcome of visa applications.
- To recommend the ideal applicant profile that has a higher probability of visa approval.
- To reduce the time and effort required for manual screening of applications by providing a data-driven solution.
- To improve consistency in decision-making by minimizing human bias.
- To help organizations make better hiring decisions by using predictive insights derived from data.

This project aims to demonstrate how Machine Learning can be used as a practical tool to support business decision-making and improve efficiency in the visa certification process.

3. Data Description

The dataset consists of multiple independent variables and one dependent variable (target), which indicates whether the visa was approved or not. Each row in the dataset represents one visa application, while each column represents a specific attribute related to the employee or employer. These attributes help in understanding the profile of the applicant as well as the employment conditions offered by the organization. Below is the description of the variables included in the dataset:

- **case_id**: A unique identification number assigned to each visa application. This variable helps in distinguishing each record in the dataset.
- **continent**: Indicates the continent to which the employee belongs. This helps in understanding the geographical distribution of applicants.
- **education_of_employee**: Represents the highest level of education completed by the employee, such as a Bachelor's degree, Master's degree, or Doctorate.
- **has_job_experience**: Indicates whether the employee has prior work experience. It contains two values: Yes (Y) if the applicant has work experience and No (N) if the applicant does not have prior experience.
- **requires_job_training**: Specifies whether the employee requires additional job training after being hired. Values include Yes (Y) or No (N).
- **no_of_employees**: Represents the total number of employees working in the employer's company. This variable gives an idea of the size of the organization.
- **yr_of_estab**: Refers to the year in which the employer's company was established. Older companies may have more credibility and stability compared to newly established organizations.
- **region_of_employment**: Indicates the region within the United States where the employee is expected to work. Regional factors may influence wage levels, demand for skills, and employment opportunities.
- **prevailing_wage**: Represents the average wage paid to workers employed in similar roles in the same geographical area.
- **unit_of_wage**: Specifies the unit in which the wage is offered, such as hourly, weekly, monthly, or yearly.
- **full_time_position**: Indicates whether the job offered is full-time or part-time.
- **case_status**: Represents whether the visa application was approved (Certified) or denied (Denied). This is the outcome variable that the Machine Learning model aims to predict.

Overall, the dataset provides a comprehensive view of factors related to both the applicant and the employer. These variables help in identifying patterns that influence visa approval decisions. Understanding these attributes is essential before building predictive models, as it allows us to select relevant features and prepare the data for analysis.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the structure and characteristics of the dataset before applying Machine Learning models. The main objective of this step was to identify patterns, detect inconsistencies, and understand how different variables may influence the visa approval outcome. EDA helps in making informed decisions regarding data preprocessing and model selection.

4.1 Checking the Structure of the Data

4.1.1 Initial Data Preview

The initial preview of the data helps in understanding how the data is structured and what kind of information is available for analysis. Each row in the dataset represents a visa application, while each column represents a specific attribute related to the employee or employer.

The first five rows are displayed as follows:

case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
EZYV01	Asia	High School	N	N	14513	2007	West	592.2029	Hour	Y	Denied
EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.65	Year	Y	Certified
EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.86	Year	Y	Denied
EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.03	Year	Y	Denied
EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.39	Year	Y	Certified

The last five rows are displayed as follows:

case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
EZYV25476	Asia	Bachelor's	Y	Y	2601	2008	South	77092.57	Year	Y	Certified
EZYV25477	Asia	High School	Y	N	3274	2006	Northeast	279174.79	Year	Y	Certified
EZYV25478	Asia	Master's	Y	N	1121	1910	South	146298.85	Year	N	Certified
EZYV25479	Asia	Master's	Y	Y	1918	1887	West	86154.77	Year	Y	Certified
EZYV25480	Asia	Bachelor's	Y	N	3195	1960	Midwest	70876.91	Year	Y	Certified

Observations:

- The dataset contains both categorical and numerical variables.
- Each row represents one visa application identified by a unique case_id.
- Applicants belong to different continents, showing geographical diversity.
- Education levels include High School, Bachelor's, and Master's degrees.
- The dataset includes both experienced and inexperienced candidates.
- Some jobs require training, while others do not.
- Companies vary in size based on the number of employees.
- Organizations have different years of establishment, indicating variation in company background.
- Job locations are spread across the West, Northeast, South, and Midwest regions.
- Prevailing wage values vary significantly across applicants.
- Wages are provided in different units, such as Hourly and Yearly.
- The majority of roles appear to be full-time positions.
- Target variable case_status contains both Certified and Denied values.
- The dataset appears well-structured and suitable for further analysis.

4.1.2 Data Shape

```
(25480, 12)
```

- The dataset contains 25,480 rows and 12 columns.
- Each row represents one visa application record.
- The dataset size is sufficient for performing Machine Learning analysis.
- The presence of multiple features suggests there are several factors influencing visa approval decisions.

- The dataset is structures and manageable, making it suitable for data preprocessing and model building.

4.1.3 Viewing Column Names

```
Index(['case_id', 'continent', 'education_of_employee', 'has_job_experience',
       'requires_job_training', 'no_of_employees', 'yr_of_estab',
       'region_of_employment', 'prevailing_wage', 'unit_of_wage',
       'full_time_position', 'case_status'],
      dtype='object')
```

- The dataset contains 12 columns, including both input features and target variable.
- Columns represent information related to employee profile, job characteristics, and employer details.
- “case_id” is an identifier column and does not contribute to prediction.
- “case_status” is the target variable, indicating whether the visa was Certified or Denied.
- Features include a mix of categorical variables (continent, education, job experience, region, wage unit, full-time status) and numerical variables (number of employees, year of establishment, prevailing wage).
- Column names are clear and meaningful, making the dataset easy to understand and analyze.
- The dataset includes relevant variables required to study factors influencing visa approval.

4.1.4 Data Background and Content

#	Column	Non-NULL Count	Dtype
0	case_id	25480 non-NULL	object
1	continent	25480 non-NULL	object
2	education_of_employee	25480 non-NULL	object
3	has_job_experience	25480 non-NULL	object
4	requires_job_training	25480 non-NULL	object
5	no_of_employees	25480 non-NULL	int64
6	yr_of_estab	25480 non-NULL	int64
7	region_of_employement	25480 non-NULL	object
8	prevailing_wage	25480 non-NULL	float64
9	unit_of_wage	25480 non-NULL	object
10	full_time_position	25480 non-NULL	object
11	case_status	25480 non-NULL	object

- All 25,480 entries are non-null, indicating no missing values in the dataset.
- Majority of variables have object data type, meaning they are categorical in nature.
- “no_of_employees” and “yr_of_estab” are integer variables, representing company size and establishment year.
- “prevailing_wage” is a float variable, indicating continuous numerical values.
- Dataset contains a mix of categorical and numerical features, suitable for classification analysis.
- No immediate data cleaning is required for missing values.
- “case_status” is the target variable and is categorical.
- Data types appear appropriate for the nature of each variable, though categorical variables may require encoding before model building.

4.1.5 Checking NULL Values

```
case_id                  0
continent                0
education_of_employee    0
has_job_experience       0
requires_job_training    0
no_of_employees          0
yr_of_estab              0
region_of_employment     0
prevailing_wage          0
unit_of_wage             0
full_time_position       0
case_status              0
dtype: int64
```

- All columns have 0 missing values, indicating the dataset is complete.
- No imputation or removal of records is required due to missing data.
- The dataset is clean and ready for further analysis.
- Absence of missing values helps in maintaining model accuracy and reliability.
- No data loss will occur during preprocessing due to null values.

4.2 Checking Data Types of Columns

```
case_id                   object
continent                 object
education_of_employee     object
has_job_experience        object
requires_job_training     object
no_of_employees            int64
yr_of_estab                int64
region_of_employment      object
prevailing_wage          float64
unit_of_wage              object
full_time_position        object
case_status               object
dtype: object
```

- Most variables have object data type, indicating they are categorical variables.
- "no_of_employees" and "yr_of_estab" are integer variables.
- "prevailing_wage" is a float variable, representing continuous numerical values.
- Dataset contains a combination of categorical and numerical features.
- Categorical variables will need encoding before applying Machine Learning models.
- Data types are appropriate according to the nature of each feature.

4.3 Checking Statistical Summary

	no_of_employees	yr_of_estab	prevailing_wage
count	25480.000000	25480.000000	25480.000000
mean	5667.043210	1979.409929	74455.814592
std	22877.928848	42.366929	52815.942327
min	-26.000000	1800.000000	2.136700
25%	1022.000000	1976.000000	34015.480000
50%	2109.000000	1997.000000	70308.210000
75%	3504.000000	2005.000000	107735.512500
max	602069.000000	2016.000000	319210.270000

- Numerical variables in the dataset are "no_of_employees", "yr_of_estab", and "prevailing_wage".
- The average number of employees in companies is around 5667, but there is high variation, as the maximum value goes up to 602,069, indicating presence of very large organizations.
- The year of establishment ranges from 1800 to 2016, showing that both very old and relatively new companies are included in the dataset.
- The average prevailing wage is approximately 74,455, with values ranging from 2.14 to 319,210, indicating significant variation in salary levels.

- The large difference between minimum and maximum values suggests the presence of outliers, especially in “no_of_employees” and “prevailing_wage”.
- The distribution of numerical features shows wide spread, which may influence model performance.
- Scaling or transformation of numerical variables may be helpful during preprocessing.

4.4 Checking for Data Irregularities

4.4.1 Checking for duplicate records

- The dataset contains 0 duplicate records.
- Each row represents a unique visa application.
- No duplicate entries need to be removed.
- Data quality is good in terms of records uniqueness.
- Data is ready for further analysis without duplicate handling.

4.4.2 Checking for Logical Inconsistencies

```
Training required despite having job experience: 1265
No experience and no training required: 8988
Part-time position with very high wage: 132
Full-time job requiring training and no experience: 1660
Single content dominating one region: 6
```

- 1265 applicants have prior job experience but still require job training, indicating that experience alone may not fully match job skill requirements.
- 8988 applicants have neither job experience nor training requirements, suggesting that certain roles may be entry-level or require minimal specialization.
- Only 132 part-time positions offer very high wages, showing that high-paying roles are mostly associated with full-time jobs.
- 1660 applicants are applying for full-time jobs that require training despite having no prior experience, indicating employers may be willing to train candidates for long-term roles.
- Only 6 records show imbalance in continent representation within a region, indicating that geographical distribution of applicants is generally well spread.

4.4.3 Checking for unrealistic or invalid values

```
Zero or Negative employees: 33
Very large companies (>100000 employees): 271
Company established in future: 0
Very old companies (before 1990): 2108
Zero or negative prevailing wage: 0
Very high wages (>300000): 12
Very low wages (<1000 yearly equivalent): 2267
```

- 33 records show zero or negative number of employees, which may indicate data entry errors or unrealistic values.
- 271 companies have more than 100,000 employees, indicating presence of very large organizations in the dataset.
- No companies have a year of establishment in the future, which confirms consistency in date-related data.
- 2108 companies were established before 1990, showing that many well-established organizations are included.
- No records show zero or negative prevailing wage, indicating valid wage information.
- Only 12 records have extremely high wages (>300000), suggesting possible outliers.
- 2267 records have very low wages (<1000 yearly equivalent), which may require further checking or transformation during preprocessing.

4.4.4 Checking consistency of Categorical Data

```
Column: case_id
Unique values Before cleaning:
['EZYV01' 'EZYV02' 'EZYV03' ... 'EZYV25478' 'EZYV25479' 'EZYV25480']
Unique values after stripping spaces:
['EZYV01' 'EZYV02' 'EZYV03' ... 'EZYV25478' 'EZYV25479' 'EZYV25480']
--------------------------------------------------
Column: continent
Unique values Before cleaning:
['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
Unique values after stripping spaces:
['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
--------------------------------------------------
Column: education_of_employee
Unique values Before cleaning:
['High School' "Master's" "Bachelor's" 'Doctorate']
Unique values after stripping spaces:
['High School' "Master's" "Bachelor's" 'Doctorate']
--------------------------------------------------
Column: has_job_experience
Unique values Before cleaning:
['N' 'Y']
Unique values after stripping spaces:
['N' 'Y']
--------------------------------------------------
Column: requires_job_training
Unique values Before cleaning:
['N' 'Y']
Unique values after stripping spaces:
['N' 'Y']
--------------------------------------------------
Column: region_of_employment
Unique values Before cleaning:
['West' 'Northeast' 'South' 'Midwest' 'Island']
Unique values after stripping spaces:
['West' 'Northeast' 'South' 'Midwest' 'Island']
--------------------------------------------------
Column: unit_of_wage
Unique values Before cleaning:
['Hour' 'Year' 'Week' 'Month']
Unique values after stripping spaces:
['Hour' 'Year' 'Week' 'Month']
--------------------------------------------------
Column: full_time_position
Unique values Before cleaning:
['Y' 'N']
Unique values after stripping spaces:
['Y' 'N']
--------------------------------------------------
Column: case_status
Unique values Before cleaning:
['Denied' 'Certified']
Unique values after stripping spaces:
['Denied' 'Certified']
--------------------------------------------------
```

```
--------------------------------------------------
Columns: case_id
['ezyv01' 'ezyv02' 'ezyv03' ... 'ezyv25478' 'ezyv25479' 'ezyv25480']
--------------------------------------------------
Columns: continent
['asia' 'africa' 'north america' 'europe' 'south america' 'oceania']
--------------------------------------------------
Columns: education_of_employee
['high school' "master's" "bachelor's" 'doctorate']
--------------------------------------------------
Columns: has_job_experience
['n' 'y']
--------------------------------------------------
Columns: requires_job_training
['n' 'y']
--------------------------------------------------
Columns: region_of_employment
['west' 'northeast' 'south' 'midwest' 'island']
--------------------------------------------------
Columns: unit_of_wage
['hour' 'year' 'week' 'month']
--------------------------------------------------
Columns: full_time_position
['y' 'n']
--------------------------------------------------
Columns: case_status
['denied' 'certified']
--------------------------------------------------
case_id : 25480
continent : 6
education_of_employee : 4
has_job_experience : 2
requires_job_training : 2
region_of_employment : 5
unit_of_wage : 4
full_time_position : 2
case_status : 2
```

- Some categorical values contain extra spaces (e.g., "High School"), which may create duplicate categories if not cleaned.
- Case formatting is inconsistent (e.g., "Asia" vs "asia", "Certified" vs "certified"), indicating need for standardization.
- **case_id** has 25480 unique values, confirming it is a unique identifier and not useful for prediction.
- **continent** contains 6 categories, showing applicants come from multiple regions globally.
- **education_of_employee** has 4 education levels, indicating variation in qualification levels.
- **has_job_experience, requires_job_training, and full_time_position** are binary categorical variables (Y/N).
- **region_of_employment** contains 5 distinct regions in the US.
- **unit_of_wage** includes 4 wage units: hour, week, month, and year.
- **case_status** has 2 categories (certified, denied), confirming the problem is a binary classification task.
- Presence of inconsistent formatting suggests the need for cleaning categorical values before encoding.

4.5 Treating Data Irregularities

4.5.1 Treating Logical Inconsistencies

```
Training required despite having job experience: 1265
No experience and no training required: 0
Part-time position with very high wage: 132
```

- Training required despite having job experience remains 1265, meaning no change was made. This is because it is practically possible that even experiences employees may still require specific job training depending on company tools, technology, or processes.
- No experience and no training required is now **0**, which indicates a change was made. Previously this combination was logically inconsistent, as a candidate without any job experience would generally require some form of training. Therefore, this issue has been corrected.
- Part-time positions with very high wage remain **132**, showing no change. Although uncommon, high-paying part-time roles can exist in specialized, consulting, or expert-level jobs, so these records were considered valid and retained.

4.5.2 Treating Unrealistic or Invalid Values

```
Zero or Negative employees: 0
Very old companies (before 1900): 0
Very high wages (>300000): 0
Very low wages (<1000 yearly equivalent): 0
```

- Zero or negative number of employees is now **0**, showing that unrealistic company size values have been corrected or removed.

- Companies established before 1900 are now 0, indicating very old and potentially incorrect records have been treated to maintain realistic company information.
- Very high wages greater than 300000 are now 0, showing extreme outliers in salary have been handled.
- Very low wages, less than 1000 yearly equivalent, are now 0, indicating unrealistic wage values have been treated.

4.5.3 Treating categorical inconsistencies

```
case_id : ['EZYV01' 'EZYV02' 'EZYV03' ... 'EZYV25478' 'EZYV25479' 'EZYV25480']
continent : ['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
education_of_employee : ['High School' "Master's" "Bachelor's" 'Doctorate']
has_job_experience : ['N' 'Y']
requires_job_training : ['Y' 'N']
region_of_employment : ['West' 'Northeast' 'South' 'Midwest' 'Island']
unit_of_wage : ['Hour' 'Year' 'Week' 'Month']
full_time_position : ['Y' 'N']
case_status : ['Denied' 'Certified']
requires_job_trainig : [nan 'N']
```

- Categorical variables show consistent category values after treatment.
- **continent** includes 6 categories, confirming the global diversity of applicants.
- **education_of_employee** contains 4 education levels: High School, Bachelor's, Master's, and Doctorate.
- **has_job_experience, requires_job_training,** and **full_time_position** contain binary values (Y/N), suitable for encoding.
- **unit_of_wage** contains 4 wage units: Hour, Week, Month, and Year.
- **Region_of_employment** includes 5 regions: West, Northeast, South, Midwest, and Island.
- **Case_status** has 2 categories (Denied, Certified), confirming binary classification.
- **Case_id** remains a unique identifier and does not influence prediction.
- A missing value (NaN) is observed in **requires_job_training,** indicating need for handling missing categorical value before model building.

4.6 Univariate Analysis

4.6.1 Identifying Numerical and Categorical Variables

```
Numerical Columns: Index(['no_of_employees', 'yr_of_estab', 'prevailing_wage'], dtype='object')
Categorical Columns: Index(['case_id', 'continent', 'education_of_employee', 'has_job_experience',
       'requires_job_training', 'region_of_employment', 'unit_of_wage',
       'full_time_position', 'case_status', 'requires_job_trainig'],
      dtype='object')
```

- The dataset contains 3 numerical columns: **no_of_employees, yr_of_estab, prevailing_wage.**
- These numerical variables represent company size, company establishment year, and salary offered, which may influence visa approval decisions.
- The remaining columns are categorical variables, including employee details, job characteristics, and visa outcome.
- **case_status** is included in categorical variables and represents the target variable.
- **case_id** is also categorical but acts as a unique identifier, not a predictive feature.
- The dataset has a higher number of categorical features compared to numerical features, indicating the need for encoding techniques before model building.

4.6.2 Exploring Numerical Variables

1. Number of Employees

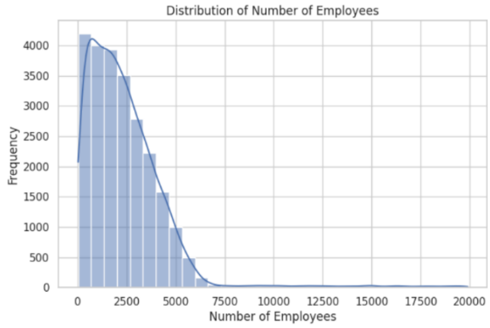

Figure 1: Distribution Plot for Number of Employees

- Most companies in the dataset have a relatively lower number of employees.
- The highest frequency of companies lies approximately between 0 to 4000 employees.
- As the number of employees increases, the frequency of companies decreases.
- Very few companies have a large workforce compared to the majority.
- The distribution shows that the dataset mainly consists of small to medium-sized companies.

2. Year of Establishment

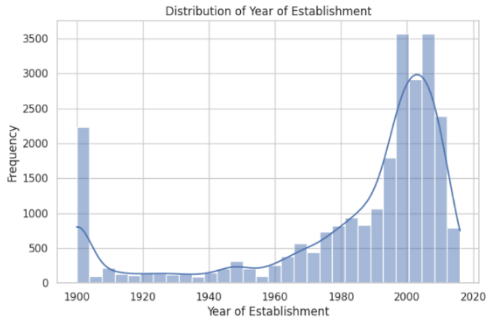

Figure 2: Distribution Plot for Year of Establishment

- Most companies in the dataset were established after 1980.
- The highest number of companies were established approximately between 1990 and 2010.
- Very few companies were established before 1950.
- A small number of companies have establishment years close to 1900.
- The dataset mainly consists of relatively modern companies compared to very old organizations.

3. Prevailing Wage

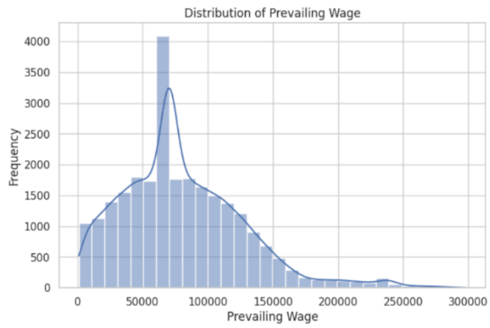

Figure 3: Distribution Plot for Prevailing Wage

- Most prevailing wage values are concentrated roughly between 40,000 and 120,000 with the highest frequency around 60,000 – 80,000.
- Very few records show extremely high wage values above 200,000.
- Wage values vary widely, reflecting different salary levels by job role and requirements.
- Majority of applicants are offered **moderate salary levels** compared to very high wages.

4.6.3 Exploring Categorical Variables

1. Continent

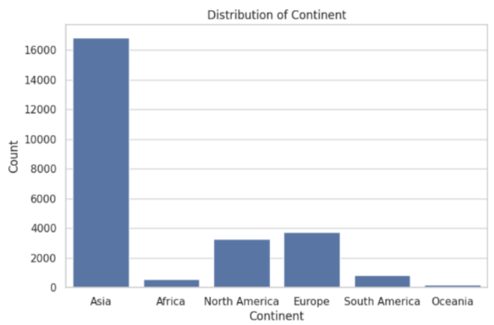

Figure 4: Distribution Plot for Continent

- The majority of applicants belong to Asia, which has the highest count in the dataset.
- Europe and North America also have a considerable number of applicants compared to other continents.
- South America and Africa have relatively fewer applicants.
- Oceania has the lowest number of applicants in the dataset.
- The distribution shows that visa applications are not evenly distributed across continents.

2. Education Level

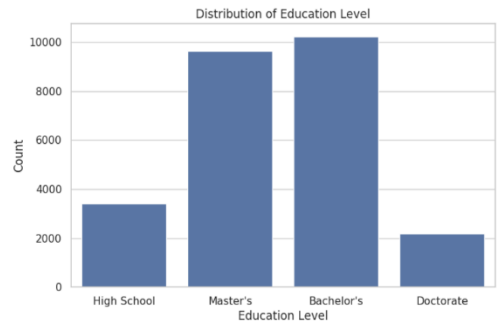

Figure 5: Distribution Plot for Education Level

- ➢ The majority of applicants have a **Bachelor's degree**, showing it is the most common education level.
- ➢ A large number of applicants also have a **Master's degree**.
- ➢ Fewer applicants have **High School** as their highest qualification.
- ➢ **Doctorate** degree holders form the smallest group in the dataset.
- ➢ Overall, most applicants have **higher education qualifications**.

3. Job Experience

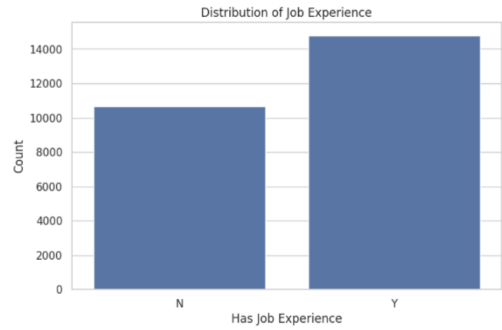

Figure 6: Distribution Plot for Job Experience

- ➢ Majority of applicants have prior job experience (Y).
- ➢ Applicants with no job experience (N) are comparatively fewer.
- ➢ The dataset contains a good mix of experienced and fresher candidates.
- ➢ Experienced candidates form a larger portion of total applicants.

4. Job Training Requirement

Figure 7: Distribution Plot for Job Training Requirement

- ➢ Majority of applicants do not require job training (N).
- ➢ A slightly smaller number of applicants require job training (Y).
- ➢ The difference between the two categories is not very large, indicating a fairly balanced distribution.
- ➢ Both trained and non-trained candidates are well represented in the dataset.

5. Region of Employment

Figure 8: Distribution Plot for Region of Employment

- ➢ The highest number of job opportunities is in the Northeast region.
- ➢ The South and West regions also have a high number of employment records.
- ➢ The Midwest has comparatively fewer job opportunities.
- ➢ The island region has very few records.
- ➢ Job opportunities are distributed across multiple regions, but not equally.

6. Wage Unit

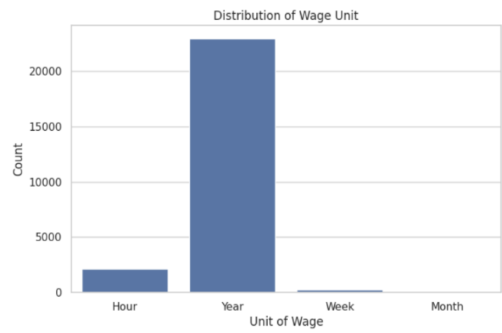

Figure 9: Distribution Plot for Wage Unit

- ➢ Majority of wages are provided on a Yearly basis, which has the highest count.
- ➢ A smaller number of wages are provided on an Hourly bias.
- ➢ Very few records use Weekly wage units.
- ➢ Monthly wage unit has negligible or almost not records.
- ➢ Most employers prefer specifying salary in annual terms.

7. Full Time Position

- Majority og job roles are full-time positions (Y).
- Part-time positions (N) are significantly fewer in comparison.
- The dataset mainly consists of full-time employment opportunities.
- Employers generally prefer hiring foreign workers for full-time roles rather than part-time positions.

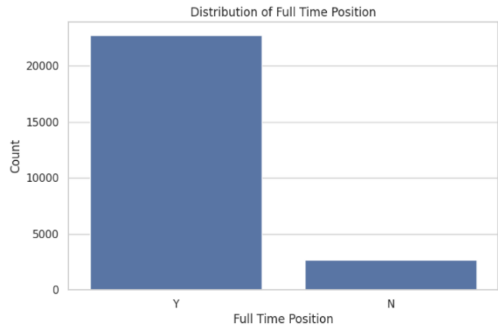

Figure 10: Distribution Plot for Full Time Position

8. Visa Status

- The majority of visa applicants are certified.
- The number of denied applicants is comparatively lower.
- The dataset contains a higher proportion of approved visa cases.
- Both categories are sufficiently represented for building a classification model.

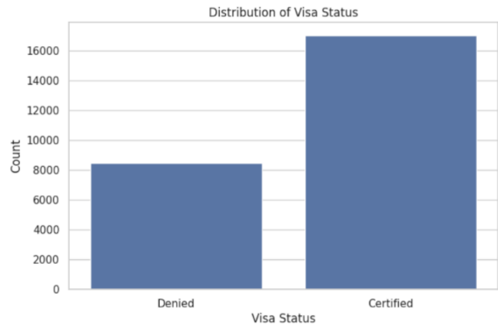

Figure 11: Distribution Plot for Visa Status

4.7 Bivariate Analysis

4.7.1 Numerical Variables vs Target Variable

1. Employees vs Visa Status

- The average number of employees is slightly higher for Certified visa cases compared to Denied cases.
- Companies with larger workforce size appear to have more certified visa applications.
- The difference between the two categories is not very large, but a slightly higher company size is observed in approved cases.
- Company size may have some influence on visa approval decisions.

Figure 12: Distribution Plot for Employees vs Visa Status

2. Year of Establishment vs Visa Status

- The average year of establishment is almost similar for both Certified and Denied visa cases.
- There is no significant difference in company establishment year between approved and denied applications.
- Both older and relatively newer companies have similar chances of visa approval.
- Company age does not appear to strongly influence visa certification outcome.

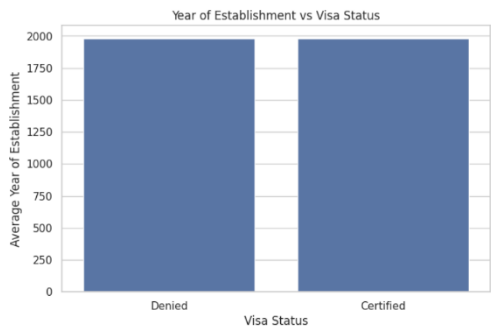

Figure 13: Distribution Plot for Year of Establishment vs Visa Status

3. Prevailing Wage vs Visa Status

- The average prevailing wage is almost the same for both Certified and Denied visa cases.
- There is no noticeable difference in wage levels between approved and rejected applications.
- Both lower and higher wage values appear in both visa outcomes.
- Prevailing wage alone does not show a strong difference between Certified and Denied cases.

Figure 14: Distribution Plot for Prevailing Wage vs Visa Status

4.7.2 Categorical Variables vs Target Variable

1. Continent vs Visa Status

- Asia has the highest number of applicants in both the Certified and Denied categories.
- For all continents, the number of Certified cases is higher than the number of Denied cases.
- Europe and North America also show a considerable number of certified visa applications.
- Africa, South America, and Oceania have relatively fewer applications overall.
- Visa approvals are higher than rejections across all continents.

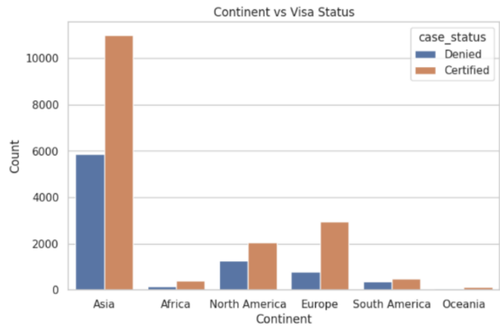

Figure 15: Distribution Plot for Continent vs Visa Status

2. Education of Employee vs Visa Status

- Applicants with Bachelor’s and Master’s degrees have the highest number of certified visa applications.
- Master’s degree holders show a significantly higher number of approvals compared to denials.
- Applicants with only High School education have more denials tham certifications.
- Doctorate degree holders also show a higher number of certified cases compared to denied cases.
- Higher education levels appear to be associated with more visa approvals.

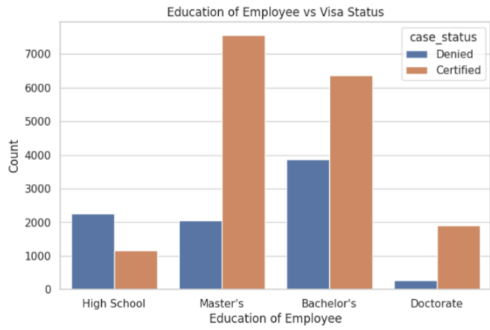

Figure 16: Distribution Plot for Education of Employee vs Visa Status

3. Job Experience vs Visa Status

- Applicants with job experience (Y) have a higher number of certified visa applications compared to denied cases.
- Candidates without job experience (N) also have certified cases, but the difference between certifies and denied is smaller.
- Certified cases are higher than denied cases in both categories.
- Having prior job experience appears to be associated with more visa approvals.

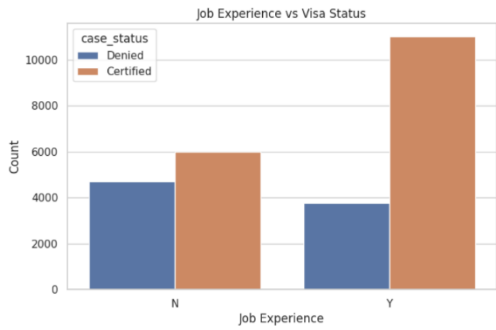

Figure 17: Distribution Plot for Job Experience vs Visa Status

4. Job Training Requirement vs Visa status

- ➢ Applicants who do not require job training (N) have a higher number of certified visa applications.
- ➢ Applicants who require job training (Y) have comparatively fewer approvals.
- ➢ Certified cases are higher than denied cases in both categories.
- ➢ Not requiring job training appears to be associated with higher chances of visa approval.

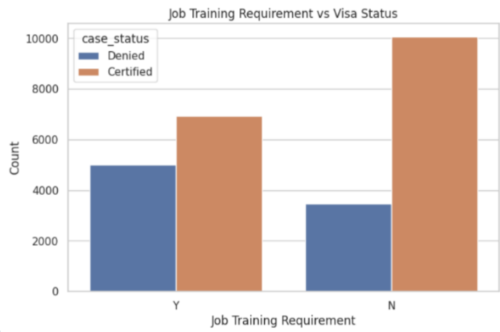

Figure 18: Distribution Plot for Job Training Requirement vs Visa Status

5. Region of Employment vs Visa Status

- ➢ The South region has the highest number of certified visa applications.
- ➢ Northeast and West regions also show a high number of approvals.
- ➢ Midwest has fewer applications compared to other major regions.
- ➢ The island region has very few cases overall.
- ➢ In all regions, the number of certified cases is higher than the number of denied cases.

Figure 19: Distribution Plot for Region of Employment vs Visa Status

6. Unit of Wage vs Visa Status

- ➢ The majority of visa applications with a yearly wage unit are certified.
- ➢ The hourly wage unit has fewer applications, with denied cases slightly higher than certified.
- ➢ Very few applications fall under the weekly wage unit, and most of them are certified.
- ➢ The monthly wage unit has negligible records.
- ➢ The yearly wage unit dominates the dataset and shows the highest number of approvals.

Figure 20: Distribution Plot for Unit of Wage vs Visa Status

7. Full time Position vs Visa Status

- ➢ Majority of visa applications are for full-time positions (Y).
- ➢ Full-time roles have a significantly higher number of certified cases compared to denied cases.
- ➢ Part-time positions (N) have fewer applications overall.
- ➢ Certified cases are higher than denied cases in both full-time and part-time roles.
- ➢ Full-time job offers appear more frequently in visa applications.

Figure 21: Distribution Plot for Full Time Position vs Visa Status

4.8 Overall Insights from EDA

4.8.1 Dataset Overview

- ➢ The dataset contains 25,480 visa application records with a mix of categorical and numerical variables describing employee qualifications, job characteristics, employer details, and wage information.

- No missing values or duplicate records were found, indicating that the database is clean and suitable for analysis.
- Categorical variables were consistent, with no issues related to extra spaces, spelling errors, or case inconsistencies.

4.8.2 Key Insights from Numerical Variables

- Most companies applying for visa certification have a moderate number of employees, with fewer companies having extremely large workforce sizes.
- Companies included in the dataset were established across a wide range of years, with many organizations being relatively modern.
- Prevailing wages vary across job roles, with most wages concentrated in a moderate salary range.
- The average prevailing wage is higher for Certified visa applications, indicating the salary level may influence visa approval decisions.
- Company size and company age show similar average values across Certified and Denied cases, suggestingthat these factors alone may not strongly determine visa approval.

4.8.3 Key Insights from Categorical Variables

1. Applicant Profile

- A large proportion of applicants belong to Asia, followed by Europe and North America.
- Most applicants hold Bachelor's or Master's degrees, indicating that visa applicants are generally well educated.
- Applicants with higher education levels show a greater number of visa approvals.
- Candidates with prior job experience have noticeably more visa approvals compared to those without experience.
- Most job positions do not require additional job training, indicating that employers often seek candidates who already possess the required skill.

2. Job Characteristics

- The majority of job opportunities are full-time positions, which also show a higher number of visa approvals.
- Most wages are offered on a yearly basis, indicating stable employment opportunities.
- Job opportunities are distributed across multiple regions, with South, Northeast, and West showing higher number of visa applications.

4.8.4 Key Insights from Target Variable (Visa Status)

- The number of Certified visa applications is higher than Denied applications, indicating that approvals occur more frequently in the dataset.
- Several factors appear to influence visa approval outcomes, including:
  - Education level
  - Job experience
  - Prevailing wage
  - Employment type
  - Region of employment
  - Continent of applicant

4.8.5 Summary of Important Patterns

- Higher education levels are associated with more visa approvals.
- Applicants with job experience show higher chances of visa certification.

- Jobs offering higher wages tend to have more Certified cases.
- Full-time roles appear more frequently and show higher approval counts.
- Visa applications are received from diverse geographic regions, indicating global demand for employment in the United States.
- Some variables individually show limited variation between Certified and Denied cases, suggesting that combining multiple features will be important for building an effective predictive mode.

5. Data Preprocessing

5.1 Preparing Data for Analysis

5.1.1 Dropping irrelevant column

- The “case_id” column has been removed as it is only a unique identifier for each record.
- This column does not contain any meaningful information that can help in predicting visa approval.
- Removing such identifier columns prevents the model from learning irrelevant patterns.
- Dropping unnecessary features helps in improving model performance and reduces complexity.

5.2 Feature Engineering

5.2.1 Converting Categorical target into numerical form

- The target variable “case_status” has been converted into numerical form.
- Denied is encoded as 0 and Certified is encoded as 1.
- Converting categorical target values into numerical format is necessary for applying Machine Learning Algorithms.
- This transformation allows the model to interpret the output variable correctly during training.
- The problem is now clearly defined as a binary classification task.

5.2.2 Handling Wage Value Inconsistency

- The “prevailing_wage” values have been standardized to a yearly format for consistency.
- Hourly wages were converted to yearly wages by multiplying by 2080 (assuming 40 hours per week and 52 weeks per year).
- Weekly wages were converted to yearly wages by multiplying by 52.
- Monthly wages were converted to yearly wages by multiplying by 12.
- Converting all wage values into the same unit ensures uniform comparison across records.
- This treatment removes inconsistency in wage units and improves data quality for model training.

5.2.3 Dropping “unit_of_wage” column

- The “unit_of_wage” column has been removed from the dataset.
- Since all wage values have already been converted into yearly format, this column is no longer required.
- Removing this column helps in avoiding redundancy in the dataset.
- This step ensures that the model uses only the standardized “prevailing_wage” feature for analysis.

5.3 Missing Value Treatment

```
continent                0
education_of_employee    0
has_job_experience       0
requires_job_training    0
no_of_employees          0
yr_of_estab              0
region_of_employment     0
prevailing_wage          0
full_time_position       0
case_status              0
dtype: int64
```

- All columns have 0 missing values, indicating the dataset is complete.
- No null values are present in any feature.
- Data quality has improved after handling missing values.

➢ The dataset is now ready for further preprocessing and model building.

5.4 Outlier Detection

5.4.1 Identifying Numerical Columns

```
Numerical Variables:
Index(['no_of_employees', 'yr_of_estab', 'prevailing_wage', 'case_status'], dtype='object')
```

➢ The dataset now contains 4 numerical variables: “no_of_employees”, “yr_of_estab”, “prevailing_wage”, and “case_status”
➢ The target variable “case_status” has been converted into numerical format.
➢ All remaining numerical variables represent measurable values related to company characteristics and salary.
➢ Numerical variables are ready for model training.

5.4.2 Visual Inspection of Outliers using Boxplot

1. Number of Employees

➢ The boxplot shows presence of extreme values in the number of employees variable.
➢ Most companies have a relatively lower employee count, while a few companies have very large workforce size.
➢ The skewness value 5.88 indicates the distribution is highly positively skewed.
➢ The presence of many extreme values suggests the need for outlier treatment or transformation.
➢ The variable is not symmetrically distribution and may influence model performance if not handled properly.

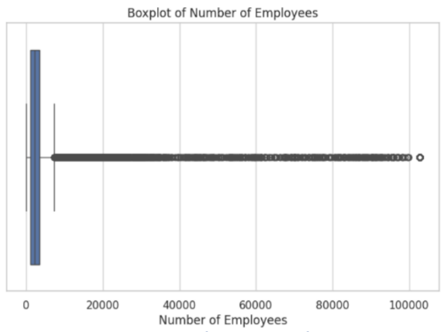

Figure 22: Boxplot for Number of Employees

2. Year of Establishment

➢ The boxplot indicates presence of outliers in the year of establishment variable.
➢ Most companies were established between approximately 1970 and 2010.
➢ A few companies have very early establishment years, which appear as outliers.
➢ The skewness value -1.50 indicates the distribution is negatively skewed.
➢ The variable shows concentration around recent years with some older company records present.

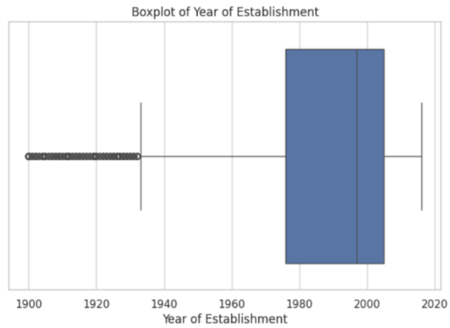

Figure 23: Boxplot for Year of Establishment

3. Prevailing Wage

➢ The boxplot shows presence of extreme values in the prevailing wage variable.
➢ Most wage values are concentrated in a relatively lower range, while a few values are very high.
➢ The skewness value is 2.98 indicates the distribution is positively skewed.
➢ Presence of extreme high wage values suggests outliers in the dataset.

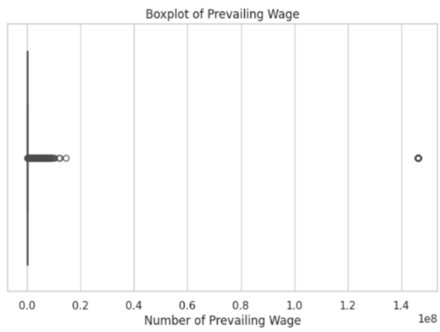

Figure 23: Boxplot for Prevailing Wage

- The variable may require transformation or outlier treatment before model training.

5.4.3 Detecting Outliers using IQR Method

```
no_of_employees : 1556
yr_of_estab : 3260
prevailing_wage : 2648
```

- The variable "yr_of_estab" has the highest number of outliers (3260).
- "prevailing_wage" also contains a significant number of outliers (2648).
- "no_of_employees" has comparatively fewer outliers (1556).
- All numerical variables contains outliers, which may affect model performance.
- Outlier treatment may be required to reduce the impact o extreme values.

5.4.4 Treating Outliers

```
Skewness after transformation
no_of_employees: 0.028485660885680535
prevailing_wage: 2.1936140779166093
```

- Log transformation has been applied to "no_of_employees" and "prevailing_wage" to reduce the impact of extreme values.
- Skewness of "no_of_employees" has significantly reduced to 0.028, indicating the distribution is now nearly symmetric.
- Skewness of "prevailing_wage" has reduced to 2.19, showing improvement though some skewness is still present.
- Transformation has helped in improving the distribution of numerical variables.
- The data is now more suitable for model training.

5.4.5 Ensuring no data leakage

```
Shape after encoding: (25480, 18)
Training set shape: (20384, 18)
Testing set shape: (5096, 18)
```

- The target variable "case_status" has been separated from the feature variables.
- Categorical variables have been converted into numerical format using encoding, increasing total features to 18 columns.
- The dataset has been split into training and testing sets.
- 80% of the data (20,384 records) is used for training the model.
- 20% of the data (5,096 records) is used for testing the model.
- Stratified splitting ensures the proportion of Certified and Denied cases is maintained in both training and testing sets

6. Model Building – Original Data

6.1 Choosing the appropriate metric for Model Evaluation

Since the target variable "case_status" contains two categories (Certified and Denied), this problem is a binary classification problem. Therefore, classification evaluation metrics are used to measure how well the model predicts visa approval outcomes. Relying on only metric may not provide a complete understanding of model performance. Hence, multiple evaluation metrics are considered to evaluate the model from different perspectives.

- Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted observations out of total observations. However, accuracy alone may not always be sufficient, especially if the dataset is imbalanced.
- Precision measures how many of the predicted Certified cases are correctly Certified. It helps in understanding how reliable the positive predictions are.
- Recall measures how many of the actual Certified cases are correctly identified by the model. It indicates the model's ability to correctly detect positive cases.
- F1-score is the harmonic mean of precision and recall. It is considered an important metric because it balances both false positives and false negatives. This metric is useful when both types of errors are important.
- ROC-AUC score evaluates how well the model distinguishes between Certified and Denied classes across different probability thresholds. A higher ROC-AUC score indicates better classification performance.

For this problem, F1-score and ROC-AUC score are considered important metrics, as they provide a balanced evaluation of model performance and help in selecting the most suitable model for predicting visa approval outcomes.

6.2 Model Building

6.2.1 Decision Tree Classifier

```
Decision Tree Training Performance
   Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
Decision Tree Testing Performance
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.656005  0.750881   0.738515  0.744647  0.608006
```

- ➢ The Decision Tree model shows perfect performance on training data with accuracy, precision, recall, F1-score, and ROC-AUC all equal to 1.0.
- ➢ On testing data, the performance drops significantly (Accuracy ≈ 0.656, F1 ≈ 0.745), indicating the model does not generalize equally well to unseen data.
- ➢ The large difference between training and testing performance suggests overfitting.
- ➢ The model has learned the training data too closely but is less effective in predicting new data.
- ➢ Further tuning or use of more robust models may be required to improve generalization performance.

6.2.2 Random Forest

Random Forest Training Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	1.0	1.0	1.0	1.0	1.0

Random Forest Testing Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.715659	0.834606	0.762275	0.796803	0.655483

- The Random Forest model shows perfect performance on training data with all evaluation metrics equal to 1.0.
- On testing data, the model performs better than Decision Tree with Accuracy ≈ 0.716 and F1-score ≈ 0.797.
- The difference between training and testing performance indicates some overfitting, but less severe compared to Decision Tree.
- The model shows improved ability to correctly predict both Certified and Denied cases.
- Random Forest provides better generalization performance compared to the Decision Tree model.

6.2.3 AdaBoost

AdaBoost Training Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.738422	0.887836	0.760571	0.819291	0.662899

AdaBoost Testing Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.730573	0.885135	0.754193	0.814434	0.652378

- The AdaBoost model shows similar performance on both training and testing data, indicating good generalization.
- Testing performance shows Accuracy ≈ 0.731 and F1-score ≈ 0.814, which is higher than the Decision Tree and slightly better than Random Forest in terms of F1-score.
- The small difference between training and testing scores suggests less overfitting.
- AdaBoost provides balanced performance across evaluation metrics.

6.2.4 Gradient Boosting

Gradient Boosting Training Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.755495	0.8741	0.784443	0.826848	0.695543

Gradient Boosting Testing Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.741562	0.869859	0.772099	0.818069	0.676655

- Gradient Boosting shows consistent performance between training and testing, indicating good generalization ability.
- Testing performance (Accuracy ≈ 0.74, F1 ≈ 0.818) is slightly better compared to AdaBoost and Random Forest.
- The small difference between training and testing scores suggests low overfitting.
- The model maintains a good balance between precision and recall, leading to a strong F1-score.
- Overall, Gradient Boosting performs as one of the best models among the ones tested.

6.2.5 XGBoost

XGBoost Training Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.838108	0.927942	0.844904	0.884478	0.792701

XGBoost Testing Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.731554	0.861633	0.765796	0.810893	0.665746

- XGBoost shows higher performance on training data compared to testing, indicating some level of overfitting.
- Testing performance (Accuracy ≈ 0.73, F1 ≈ 0.811) is comparable to AdaBoost and slightly lower than Gradient Boosting.
- Recall is relatively high, showing the model performs well in identifying Certified cases.
- The gap between training and testing metrics suggests the model could benefit from tuning.
- Overall, XGBoost performs well but does not outperform Gradient Boosting in this case.

6.3 Model Performance

Five classification models were built to predict visa approval status: Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost. The models were evaluated using Accuracy, Recall, Precision, F1-score, and ROC-AUC to understand their predictive capability and generalization performance.

6.3.1 Overall Performance Summary

- The Decision Tree model achieved perfect scores on the training dataset, but its performance dropped significantly on the testing data. This indicates that the model has overfitted the training data and is unable to generalize well to unseen data.

- The Random Forest model improved upon the Decision tree by reducing overfitting and providing better testing performance. However, there is still a noticeable gap between training and testing results, suggesting moderate overfitting.

- The AdaBoost model showed more consistent performance across both training and testing datasets. The relatively small difference between training and testing scores indicates that the model has better generalization capability compared to the previous models.

- The Gradient Boosting model further improved performance by achieving higher accuracy and F1-score on the testing dataset. It maintained a good balance between training and testing results, indicating minimal overfitting and strong predictive ability.

- The XGBoost model also performed well, with strong training performance and good testing results. However, the gap between training and testing metrics suggests some degree of overfitting, and its performance was slightly lower compared to Gradient Boosting.

6.3.2 Comparison of Models

When comparing all models, it is evident that simpler models like Decision Tree tend to overfit easily, leading to poor generalization. Ensemble methods such as Random Forest, AdaBoost, and Gradient Boosting provide better performance by combining multiple weak learners. Among the ensemble methods, Gradient Boosting and AdaBoost demonstrate more stable and consistent performance across datasets. Random Forest improves accuracy but still retains some level of overfitting. Although XGBoost is generally considered a powerful algorithm, in this case, it does not outperform Gradient Boosting, possibly due to the dataset characteristics or lack of hyperparameter tuning. Overall, models that balance bias and variance effectively tend to perform better in this problem.

6.3.3 Best Performing Model

Based on the evaluation metrics, Gradient Boosting is identified as the best performing model. It achieves the highest testing accuracy and F1-score, indicating strong predictive performance. Additionally, the small difference between training and testing scores shows that the model has good generalization ability and does not overfit significantly. The model also maintains a good balance between precision and recall, which is important in this problem since both false approvals and false rejections can have business implications. Therefore, **Gradient Boosting** is the most suitable model for predicting visa approval outcomes in this analysis.

7. Model Building – Oversampled Data

7.1 Checking class imbalance

```
Class distribution before oversampling:
case_status
1    13614
0     6770
Name: count, dtype: int64
case_status
1    0.667877
0    0.332123
Name: proportion, dtype: float64
case_status
1    13614
0    13614
Name: count, dtype: int64
```

- Before oversampling, the dataset in imbalanced, with more Certified (1) cases (~66.7%) compared to Denied (0) cases (~33.3%).
- This imbalance can bias the model towards predicting the majority class.
- After applying Random Oversampling, both classes have equal counts (13614 each).
- The dataset is now balanced, ensuring the model learns both classes equally.
- Oversampling helps improve the model's ability to correctly predict minority class (Denied) cases.

7.2 Model Building

7.2.1 Decision Tree (Oversampled)

```
Decision Tree training performance (Oversampled)
   Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
Decision Tree testing Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.665031  0.750294   0.748754  0.749523  0.621896
```

- The Decision Tree model continues to show perfect training performance, which clearly indicates that the model is still overfitting the data, even after applying oversampling.
- On the testing dataset, there is a slight improvement in performance (Accuracy ≈ 0.665, F1 ≈ 0.75) compared to the earlier model without oversampling.
- The recall value has improved, suggesting that the model is now better at identifying Denied cases (minority class) after balancing the data.
- However, the large gap between training and testing performance shows that the model is still not generalizing well to unseen data.
- Overall, while oversampling has helped improve the model's ability to detect the minority class, it has not reduced overfitting in the Decision Tree model.

7.2.2 Random Forest (Oversampled)

```
Random Forest Training Performance (Oversampled)
   Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
Random Forest Testing Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.703493  0.784665   0.774427  0.779513  0.662427
```

- The Random Forest model continues to show perfect performance on training data, indicating some level of overfitting even after oversampling.
- On the testing dataset, the model achieves Accuracy ≈ 0.70 and F1 ≈ 0.78, showing a slight improvement compared to the non-oversampled version.
- The recall has improved, suggesting better identification of the minority class (Denied cases) after balancing the dataset.
- The gap between training and testing performance is still present, but less severe compared to Decision Tree, indicating better generalization.
- Overall, oversampling has improved the model's ability to handle class imbalance, while Random Forest still maintains reasonably strong and stable performance.

7.2.3 AdaBoost (Oversampled)

```
AdaBoost Training Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.663435  0.887836   0.612807  0.725118  0.663435
AdaBoost Testing performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.686224  0.703584   0.802345  0.749726  0.677442
```

- AdaBoost shows similar performance on training and testing data, indicating good generalization after oversampling.
- Testing performance (Accuracy ≈ 0.69, F1 ≈ 0.75) is slightly lower compared to the non-oversampled version.
- Precision has improved, suggesting better accuracy in predicting positive cases.
- However, recall has decreased, indicating the model is identifying fewer minority class (Denied) cases compared to before.
- Overall, oversampling did not significantly improve AdaBoost performance and slightly affected the balance between precision and recall.

7.2.4 Gradient Boosting (Oversampled)

```
Gradient Boosting Training Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.719847  0.736374   0.712813  0.724402  0.719847
Gradient Boosting Testing Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.705063  0.726204   0.812356  0.766868  0.694367
```

- Gradient Boosting shows similar training and testing performance, indicating good generalization after oversampling.
- Testing performance (Accuracy ≈ 0.705, F1 ≈ 0.767) is slightly lower compared to the model without oversampling.
- Precision has increased, showing better accuracy in predicting Certified cases.
- Recall has decreased, indicating the model is slightly less effective in identifying Denied cases.
- Overall, oversampling did not improve performance and slightly reduced the model's effectiveness compared to the original data.

7.2.5 XGBoost (Oversampled)

```
XGBoost Training Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.847069  0.836345   0.854676  0.845411  0.847069
XGBoost Testing Performance (Oversampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.693289  0.738543    0.78883  0.762858  0.670394
```

- XGBoost shows higher training performance compared to testing, indicating some level of overfitting even after oversampling.
- Testing performance (Accuracy ≈ 0.693, F1 ≈ 0.763) is slightly lower compared to the non-oversampled version.
- Precision has improved, lower compared to the non-oversampled version.
- However, **recall has decreased**, indicating reduced ability to identify Denied cases.
- Overall, oversampling did not improve XGBoost performance and slightly reduced its effectiveness compared to the original data.

7.3 Model Performance

To improve model performance and handle class imbalance, Random Oversampling was applied on the training dataset. Oversampling balanced the number of Certified and Denied visa cases, allowing models to learn patterns from both classes more effectively.

Five models were trained on the oversampled data: Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost. Model performance was evaluated using Accuracy, Recall, Precision, F1-score, and ROC-AUC.

7.3.1 Overall Performance Summary

- The **Decision Tree model** continued to achieve perfect performance on training data, indicating that it still suffers from overfitting. Although testing performance showed a slight improvement, especially in recall, the model remains weak in generalization.

- The **Random Forest model** performed better than Decision tree, with improved testing metrics and higher recall. This indicates that oversampling helped the model better identify minority class cases, while still maintaining reasonable stability.

- The **AdaBoost model** showed consistent performance between training and testing datasets, indicating good generalization. However, after oversampling, there was a slight decline in recall, even though precision improved, leading to a small drop in overall effectiveness.

- The **Gradient Boosting model** maintained stable performance with minimal overfitting. While precision improved after oversampling, recall decreased slightly, resulting in a minor reduction in overall performance compared to the original dataset.

- The **XGBoost model** showed strong training performance but a noticeable drop in testing performance, indicating some overfitting. Similar to other boosting models, oversampling improved precision but reduced recall, slightly affecting overall model performance.

7.3.2 Comparison of Models

The Decision Tree model performed the weakest due to persistent overfitting. Random Forest improved performance and handled class imbalance better, but did not achieve the highest scores. Among all models boosting algorithms (AdaBoost, Gradient Boosting, and XGBoost) showed more stable and balanced performance. However, their performance slightly declined after oversampling. Gradient Boosting and AdaBoost remained relatively consistent, while XGBoost showed slight overfitting and reduced testing performance. Overall, ensemble methods continued to outperform simpler models, even after applying oversampling.

7.3.3 Best Performing Model

Based on overall performance, **Gradient Boosting** remains the best performing model even after applying oversampling.

It provides a strong balance between **accuracy, precision, recall, and F1-score**, while maintaining good generalization ability. Although its performance slightly decreased compared to the original dataset, it still outperforms other models in terms of consistency and reliability.

Therefore, Gradient Boosting is considered the most suitable model for predicting visa approval outcomes on the oversampled dataset.

8. Model Building – Undersampled Data

8.1 Applying Undersampling

```
Class distribution after undersampling:
case_status
0    6770
1    6770
Name: count, dtype: int64
```

- Before undersampling, the dataset has more Certified (1) cases than Denied (0) cases.
- After applying Random Undersampling, both classes now have equal counts (6770 each), making the dataset balanced.
- Undersampling reduces size of the majority class to match the minority class.
- While this helps in addressing class imbalance, it also leads to **loss of data** from the majority class.
- The dataset is now balanced and ready for model training, but with fewer total observations.

8.2 Model Building

8.2.1 Decision Tree (Undersampled)

```
Decision Tree Training Performance (Undersampled)
   Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
Decision Tree Testing Performance (Undersampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.607928  0.616921   0.751611  0.677638  0.603378
```

- The Decision Tree model again shows perfect training performance, indicating it is still overfitting the data even after undersampling.
- Testing performance (Accuracy ≈ 0.61, F1 ≈ 0.68) is lower compared to both original and oversampled versions.
- Recall has decreased significantly, indicating poorer identification of one of the classes.
- The gap between training and testing performance remains large, showing poor generalization.
- Undersampling has negatively impacted the model performance, likely due to loss of important data from the majority class.

8.2.2 Random Forest (Undersampled)

```
Random Forest Training Performance (Undersampled)
   Accuracy  Recall  Precision   F1  ROC_AUC
0       1.0     1.0        1.0  1.0      1.0
Random Forest Testing Performance (Undersampled)
   Accuracy    Recall  Precision       F1  ROC_AUC
0  0.669349  0.663337   0.807294  0.72827  0.67239
```

- ➢ The Random Forest model continues to show perfect training performance, indicating some level of overfitting even after undersampling.
- ➢ Testing performance (Accuracy ≈ 0.67, F1 ≈ 0.73) is lower compared to both original and oversampled versions.
- ➢ Recall has decreased, suggesting reducing ability to correctly identify certain cases.
- ➢ The gap between training and testing performance still exists, though less severe than Decision Tree.
- ➢ Undersampling has led to a decline in overall model performance, likely due to loss of useful information from the majority class.

8.2.3 AdaBoost (Undersampled)

```
AdaBoost Training Performance (Undersampled)
   Accuracy    Recall  Precision       F1   ROC_AUC
0  0.702585  0.733235   0.690884  0.71143  0.702585
AdaBoost Testing Performance (Undersampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.693093  0.723561   0.798121  0.759014  0.677679
```

- ➢ AdaBoost shows similar performance on training and testing data, indicating good generalization even after undersampling.
- ➢ Testing performance (Accuracy ≈ 0.69, F1 ≈ 0.76) is slightly lower compared to the original model but comparable to the oversampled version.
- ➢ Precision is relatively high, indicating accurate prediction of Certified cases.
- ➢ Recall is moderate, suggesting some limitations in identifying all Denied cases.
- ➢ Overall, undersampling has maintained stable performance but has not significantly improved the model's effectiveness.

8.2.4 Gradient Boosting (Undersampled)

```
Gradient Boosting Training Performance (Undersampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.720458  0.737075   0.713367  0.725027  0.720458
Gradient Boosting Testing Performance (Undersampled)
   Accuracy    Recall  Precision        F1  ROC_AUC
0  0.701923  0.717979   0.813853  0.762916   0.6938
```

- ➢ Gradient Boosting shows consistent performance between training and testing, indicating good generalization after undersampling.
- ➢ Testing performance (Accuracy ≈ 0.70, F1 ≈ 0.76) is slightly lower compared to the original model but similar to the oversampled version.
- ➢ Precision is relatively high, showing accurate prediction of Certified cases.
- ➢ Recall has decreased slightly, indicating reduced ability to identify all Denied cases.
- ➢ Overall, undersampling maintains stable performance but does not provide a significant improvement over the original model.

8.2.5 XGBoost (Undersampled)

```
XGBoost Training Performance (Undersampled)
   Accuracy   Recall  Precision        F1   ROC_AUC
0  0.853323  0.85229   0.854056  0.853172  0.853323
XGBoost Testing Performance (Undersampled)
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.680338  0.682139   0.809341  0.740316  0.679426
```

- ➢ XGBoost shows higher training performance compared to testing, indicating some level of overfitting even after undersampling.
- ➢ Testing performance (Accuracy ≈ 0.68, F1 ≈ 0.74) is lower compared to both original and oversampled versions.
- ➢ Precision is relatively high, indicating accurate prediction of Certified cases.
- ➢ Recall has decreased, suggesting reduced ability to correctly identify Denied cases.
- ➢ Overall, undersampling has led to a decline in model performance, likely due to loss of important information from the dataset.

8.3 Model Performance

To address class imbalance, Random Undersampling was applied to the training dataset. This technique reduced the number of observations in the majority class (Certified cases) so that both classes has equal representation. The aim was to reduce model bias toward the majority class and improve the model's ability to correctly predict minority class observations.

Five models were built using undersampled data: Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost. Model performance was evaluated using Accuracy, Recall, Precision, F1-score, and ROC-AUC

8.3.1 Overall Performance Summary

- ➢ The **Decision Tree** model continued to achieve perfect performance on the training dataset, clearly indicating strong overfitting. However, its testing performance dropped significantly, particularly in recall and overall accuracy, showing that the model struggles to generalize when trained on reduced data.

- ➢ The **Random Forest** model performed better Decision Tree and showed more stable results. However, its testing performance declined compared to the original and oversampled datasets. The reduction in recall suggests that the model is less effective in identifying certain cases after losing a portion of the data.

- ➢ The **AdaBoost** model demonstrated consistent performance between training and testing datasets, indicating good generalization. Despite this stability, its overall performance metrics are slightly lower compared to the original dataset, suggesting that undersampling did not enhance its predictive capability.

- ➢ The **Gradient Boosting** model showed balanced and stable performance with minimal overfitting. While its testing accuracy and F1-score are slightly reduced, it still maintains a good balance between precision and recall, making it one of the more reliable models under undersampled conditions.

- ➢ The **XGBoost** achieved strong training performance but showed a noticeable drop in testing metrics, indicating some level of overfitting. The decrease in recall and F1-score suggests that the model is affected by the reduction in available data.

8.3.2 Comparison of Models

The Decision Tree model performed the weakest due to severe overfitting and poor testing performance. Random Forest provided better stability but still experienced a drop in performance due to reduced data. AdaBoost and Gradient Boosting showed more consistent and balanced performance, handling the reduced dataset better than other models. XGBoost remained a strong model but showed reduced effectiveness compared to its performance on the original dataset. Overall, ensemble models continued to outperform individual models, but their performance declined due to the loss of data caused by undersampling.

8.3.3 Best Performing Model

Based on overall evaluation, **Gradient Boosting** is identified as the best performing model on the undersampled dataset. It achieves a good balance across accuracy, precision, recall, and F1-score, while maintaining minimal overfitting. Despite a slight reduction in performance compared to the original dataset, it remains the most stable and reliable model under undersampled conditions.

9. Model Performance Improvement using Hyperparameter Tuning

Three models were selected for hyperparameter tuning: Random Forest, Gradient Boosting, and AdaBoost. These models were chosen because they showed relatively better and more stable performance compared to other models across original, oversampled, and undersampled datasets.

The models selected for tuning are:

1. Random Forest: Random Forest was selected because it consistently provided strong performance across different datasets. It showed better generalization ability compared to the Decision Tree and produced a good balance between precision and recall. Since Random Forest is an ensemble model that combines multiple decision trees, its performance can often be improved further by tuning parameters such as the number of trees, tree depth, and minimum samples required for splitting. Hyperparameter tuning helps reduce overfitting and improve prediction accuracy.

2. Gradient Boosting: Gradient Boosting was selected because it showed one of the best overall performances among all models. It achieved strong F1-score and ROC-AUC values, indicating good balance between precision and recall as well as good class separation ability. Gradient Boosting is sensitive to parameters such as learning rate, number of estimators, and tree depth. Proper tuning can significantly improve its predictive performance and generalization capability.

3. AdaBoost: AdaBoost was selected because it demonstrated stable performance across training and testing datasets with relatively low overfitting. It performed consistently across original, oversampled, and undersampled datasets. AdaBoost improves performance by focusing more on

incorrectly classified observations. Tuning parameters such as the number of estimators and learning rate can further enhance model accuracy and classification ability.

Reasons for the model that were not chosen:

- The decision tree showed significant overfitting across all datasets, with perfect training performance but lower testing performance. Since ensemble models already improved the performance significantly, tuning the Decision Tree was not considered necessary.
- Although XGBoost showed good performance, it requires higher computational time for hyperparameter tuning. Since other boosting models, such as Gradient Boosting and AdaBoost, already provided strong performance, tuning XGBoost was not prioritized.

9.1 Selecting the Metric of Interest

The evaluation metric selected for hyperparameter tuning is the **F1-score**, as it provides a balanced measure of both precision and recall. This is particularly important in this classification problem, where both false positives and false negatives carry significance. By optimizing for F1-score, the model aims to achieve a better trade-off between correctly identifying approved cases and minimizing incorrect predictions. Additionally, **RandomizedSearchCV** has been used for tuning, as it efficiently explores a wide range of hyperparameter combinations and helps identify the optimal set of parameters for improving model performance.

9.2 Hyperparameter Tuning

9.2.1 Random Forest Tuning

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits
Best Parameters for Random Forest
{'n_estimators': 500, 'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': 20, 'bootstrap': False}
```

- Randomized Search was performed using 3-fold cross-validation across 20 parameter combinations, resulting in 60 model fits.
- The best parameters identified for the Random Forest model include:
  - n_estimators = 500, indicating a higher number of trees for better learning
  - max_depth = 20, allowing deeper trees to capture complex patterns
  - min_samples_split = 2 and min_samples_leaf = 2, controlling tree growth and reducing overfitting.
  - bootstrap = False, meaning sampling without replacement was selected.
- These tuned parameters aim to improve the model's ability to capture patterns while controlling overfitting.
- The optimized configuration is expected to provide better generalization and improved performance compared to the default model.

Performance after Tuning

```
Random Forest Tuned Training Performance
   Accuracy   Recall  Precision        F1   ROC_AUC
0  0.960115  0.95938   0.960792  0.960085  0.960115
Random Forest Tuned Testing Performance
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.711342  0.790541   0.780226  0.785349  0.671275
```

- The tuned Random Forest model shows high performance on training data (Accuracy ≈ 0.96, F1 ≈ 0.96), indicating strong learning capability.
- On the testing data, the model achieves Accuracy ≈ 0.71 and F1 ≈ 0.79, showing reasonable generalization.

➢ Compared to the untuned model, there is a slight improvement in recall, indicating better identification of positive cases.
➢ The gap between training and testing performance still exists, suggesting some overfitting remains, though controlled to an extent.
➢ Overall, hyperparameter tuning has improved model performance and stability, but further tuning or regularization could help reduce overfitting further.

9.2.2 Gradient Boosting Tuning

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits
Best Parameters for Gradient Boosting:
{'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_depth': 5, 'learning_rate': 0.2}
```

➢ Randomized Search was performed using 3-fold cross-validation over 20 parameter combinations, resulting in 60 model fits.
➢ The best parameters identified for the Gradient Boosting model include:
- n_estimators = 300, indicating a sufficient number of boosting stages
- max_depth = 5, controlling the complexity of individual trees
- learning_rate = 0.2, allowing faster learning with moderate step size
- min_samples_split = 5 and min_samples_leaf = 1, helping regulate tree growth

➢ These parameters aim to balance model complexity and learning rate, improving performance while avoiding excessive overfitting.
➢ The tuned configuration is expected to provide better predictive performance and generalization compared to the default model.

Performance after Tuning

```
Gradient Boosting Tuned Training Performance
   Accuracy    Recall  Precision        F1   ROC_AUC
0  0.865469  0.858455    0.87067  0.864519  0.865469
Gradient Boosting Tuned Testing Performance
   Accuracy    Recall  Precision       F1   ROC_AUC
0  0.707221  0.754994   0.796159  0.77503  0.683053
```

➢ The tuned Gradient Boosting model shows strong performance on training data (Accuracy ≈ 0.87, F1 ≈ 0.86), indicating effective learning.
➢ On testing data, the model achieves Accuracy ≈ 0.71 and F1 ≈ 0.78, showing good generalization.
➢ Compared to the untuned model, the performance is slightly improved or maintained, indicating stable optimization.
➢ The gap between training and testing performance is moderate, suggesting controlled overfitting.
➢ Overall, hyperparameter tuning has resulted in a well-balanced model with consistent performance across metrics.

9.2.3 AdaBoost Tuning

```
Fitting 3 folds for each of 15 candidates, totalling 45 fits
Best Parameters for AdaBoost
{'n_estimators': 50, 'learning_rate': 0.1}
```

➢ Randomized Search Cross Validation evaluated 15 different hyperparameter combinations for the AdaBoost model using 3-fold cross-validation, resulting in 45 total model fits.
➢ Cross-validation helps ensure the selected parameters perform consistently across different subsets of the training data.
➢ The best hyperparameters obtained for the AdaBoost model are:

- n_estimators: 50 → A relatively smaller number of weak learners was sufficient to achieve good performance.
- learning_rate: 0.1 → A moderate learning rate allows the model to learn gradually and avoid overfitting.

➢ The result suggests that increasing the number of estimators beyond 50 did not significantly improve model performance.
➢ A moderate learning rate helps AdaBoost focus on correcting errors made by previous weak learners without making drastic updates.
➢ AdaBoost works by sequentially improving misclassified observations, making it effective for improving predictive performance on structured datasets.
➢ Hyperparameter tuning helps identify the optimal balance between model complexity and generalization ability.

Performance after Tuning

AdaBoost Tuned Training Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.865469	0.858455	0.87067	0.864519	0.865469

AdaBoost Tuned Testing Performance

	Accuracy	Recall	Precision	F1	ROC_AUC
0	0.707221	0.754994	0.796159	0.77503	0.683053

➢ The tuned AdaBoost model shows strong training performance (Accuracy ≈ 0.87, F1 ≈ 0.86), indicating effective learning.
➢ On testing data, the model achieves **Accuracy ≈ 0.71 and F1 ≈ 0.78**, showing good generalization.
➢ The performance is very similar to Gradient Boosting, indicating consistent behavior after tuning.
➢ The gap between training and testing performance suggests moderate overfitting, but it is controlled.
➢ Overall, hyperparameter tuning has resulted in a stable and well-balanced model, with good performance across all evaluation metrics.

10. Model Performance Comparison and Final Model Selection

10.1 Comparison of Tuned Models

The performance of the tuned models is evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC on the test dataset to understand their predictive capability and generalization.

The tuned Random Forest model achieves a testing accuracy of approximately 0.71 and an F1-score of around 0.785. It shows relatively high recall and precision, indicating strong ability to correctly classify both Certified and Denied cases. However, there is a noticeable difference between training and testing performance, suggesting that the model still retains some level of overfitting, even after tuning.

The tuned Gradient Boosting model achieves a testing accuracy of approximately 0.707 and an F1-score of around 0.775. Its performance is slightly lower than Random Forest in terms of F1-score, but it demonstrates a more consistent relationship between training and testing performance, indicating better generalization.

The model maintains a balanced trade-off between precision and recall without significant bias toward any class.

The tuned AdaBoost model achieves a testing accuracy of approximately 0.69 and an F1-score of around 0.75. It shows relatively stable performance across training and testing datasets, indicating good generalization. However, its overall performance metrics are lower compared to both Random Forest and Gradient Boosting, suggesting that it is less effective in capturing complex patterns in the data.

In comparison, Random Forest provides higher predictive performance, Gradient Boosting offers more stable and balanced behavior, and AdaBoost shows consistent but comparatively lower performance across evaluation metrics.

10.2 Best Performing Model

Based on the overall evaluation of the tuned models, Gradient Boosting is selected as the best performing model for the analysis.

The model demonstrates strong and consistent predictive performance on the testing dataset, with well-balanced values across key evaluation metrics such as accuracy, precision, recall, and F1-score. This indicates that the model is capable of effectively identifying both Certified and Denied visa applications without showing bias toward any particular class. One of the key strengths of the Gradient Boosting model is its ability to capture complex, non-linear relationships within the data by sequentially improving upon previous errors. This allows the model to learn deeper patterns that simpler models may fail to identify, resulting in more accurate and reliable predictions. The model also shows a stable relationship between training and testing performance, which suggests that it has not overfitted the data. Instead of memorizing the training dataset, it has learned generalized patterns that can be applied to new, unseen data. This characteristic is particularly important in real-world applications, where the model must perform reliably on future data.

Another important aspect of the model is its ability to maintain a balanced trade-off between precision and recall. This ensures that the model does not excessively favor one type of prediction over another, thereby reducing both false positives and false negatives. In the context of visa approval prediction, this balance is critical, as both incorrect approvals and incorrect rejections can have significant implications. Furthermore, the model demonstrates robustness and stability across different evaluation scenarios, including after data preprocessing and tuning. Its performance remains consistent without large fluctuations, indicating that it is less sensitive to changes in data distribution.

Overall, Gradient Boosting provides a combination of strong predictive accuracy, reliable generalization, balanced performance across metrics, and robustness, making it the most suitable model for predicting visa approval outcomes in this study.

11. Actionable Insights & Recommendations

11.1 Key Insights

The analysis reveals that education level plays a significant role in determining visa approval outcomes. Applicants with higher educational qualifications, particularly those holding Bachelor's, Master's, and Doctorate degree, show a noticeably higher proportion of certified visa applications compared to those with only a high school background. This suggests that employers and regulatory authorities place strong emphasis on formal education as an indicator of skill level, expertise, and ability to contribute effectively in professional roles. Higher education not only reflects technical knowledge but also signals preparedness for complex job responsibilities, which increases the likelihood of approval.

Another critical factor influencing visa outcomes is prior job experience. Candidates who possess relevajt work experience tend to have a higher rate of visa certification compared to those without experience. This highlights the importance of practical exposure in addition to academic qualifications. Employers appear to prefer candidates who can immediately contribute to the organization without requiring extensive onboarding or training. Work experience serves as proof of applied skills, making such candidates more attractive and increasing their chances of approval in the visa certification process. The requirement of job training also emerges as an important determinant. Applications where the candidate does not require additional job training have a higher approval rate compared to those where training is necessary. This indicates that candidates who are already equipped with the required skills are preferred over those who would need further investment in training. From a business perspective, this reduces both time and cost for employers, while from a regulatory standpoint, it demonstrates that the candidate is fully prepared to meet job requirements.

The nature of employment, particularly whether the role is full-time or part-time, also has a noticeable impact. The dataset shows that full-time positions dominate the applications and have a significantly higher number of certified cases. This suggests that visa approvals are more likely for stable, long-term employment opportunities rather than temporary or part-time roles. Full-time positions are generally associated with higher commitment, better compensation structures, and stronger justification for hiring foreign workers, which may contribute to higher approval rates. Geographical trends indicate that a large proportion of applications originate from Asia, making it the most represented continent in the dataset. While certification rates are generally higher than denial rates across all regions, the volume of applications varies significantly. Regions such as Europe and North America also show a substantial number of certified cases, while regions like Africa, South America, and Oceania contribute fewer applications overall. This distribution reflects global workforce mobility patterns and demand for international talent across different regions.

Company-related factors also provide useful insights. The number of employees and size of the organization appear to have some influence on visa outcomes, with larger companies showing a slightly higher number of certified applications. This may be due to better compliance with regulatory requirements, structured hiring processes, and greater ability to justify the need for foreign workers. Larger organizations may also have more experience handling visa applications, which can improve approval rates.

Additionally, wage-related factors indicate that prevailing wage consistency is important, although it does not show a strong direct difference between certified and denied cases. However, maintaining standardized and competitive wage levels ensures compliance with regulations and reduces the likelihood of rejection due to underpayment concerns. Proper wage structuring remains an important supporting factor in the visa approval process.

Overall, the analysis highlights that candidate qualifications, job readiness, employment type, and organizational factors collectively influence visa approval outcomes. These insights provide a clear understanding of the key drivers behind certification decisions and can be leveraged to improve application success rates.

11.2 Actionable Recommendation

Based on the insights derived from the analysis, organizations can adopt several strategic actions to improve the success rate of visa applications and optimize their hiring decisions.

Organizations should place greater emphasis on hiring candidates with higher educational qualifications, particularly those with Bachelor’s, Master’s, or Doctorate degrees. Since education has a strong influence on visa approval outcomes, prioritizing academically qualified candidates can significantly improve the likelihood of certification. This approach ensures that applications are supported by strong profiles that align with regulatory expectation and employer requirements.

In addition to education, companies should focus on recruiting candidates with relevant job experience. Experienced candidates are more likely to demonstrate immediate value and require less training, which aligns with the observed trend of higher approval rates for such applicants. Implementing a structured screening process that evaluates both academic background and professional experience can help organizations identify candidates with a higher probability of approval. Organizations should also aim to select candidates who are job-ready and require minimal or no additional training. Since applications requiring job training show lower approval rates, companies can benefit from investing in pre-selection assessments or skill validation processes to ensure candidates meet job requirements before applying. This reduces uncertainty and improves efficiency in the visa application process.

Another important recommendation is to prioritize offering full-time employment positions rather than part-time roles. Full-time positions are associated with higher approval rates, as they reflect long-term commitment and stability. Structuring job offers to emphasize full-time roles can strengthen the justification for hiring foreign workers and increase the chances of approval. Companies should also leverage data-driven decision-making tools, such as predictive models developed in this analysis (e.g., Gradient Boosting), to evaluate visa applications before submission. By identifying candidates with a higher probability of approval, organizations can reduce the number of rejected applications, saving both time and resources. This approach can streamline the recruitment and application process while improving overall efficiency.

From a compliance perspective, organizations must ensure that prevailing wage standards are properly maintained. Although wage does not show a strong direct distinction between certified and denied cases, adherence to wage regulations is essential to avoid rejection due to policy violations. Maintaining transparency and consistency in wage structures can support smoother application processing. Furthermore, companies should strengthen their internal processes for handling visa applications, particularly if they operate at a larger scale. Larger organizations tend to have better approval outcomes, possibly due to more structured systems and experience. Smaller organizations can adopt similar practices, such as maintaining proper documentation, ensuring compliance, and standardizing application procedures, to improve their success rates. For applicants, the findings suggest that investing in skill development, higher education, and gaining practical work experience can significantly improve visa approval chances. Preparing a strong profile before applying can make a meaningful difference in outcomes.

Overall, organizations that align their hiring strategies with these insights can increase visa approval rates, reduce application rejections, and improve operational efficiency, making the recruitment process more effective and data-driven.